

Monte-Carlo Simulation for Checking Conjectures

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We want to check the conjecture that the stochastic fixed-point equation

$$X \stackrel{d}{=} I(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)}) \quad (1.1)$$

has a unique solution (α^*, X_{α^*}) such that $\mathbb{E}[X_{\alpha^*}] = 1$. Let T_α be the operator defined by the right-hand side of the equation, i.e.,

$$T_\alpha(\mu) = I(U^\alpha X^{(1)} + (1 - U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1 - U)^\alpha X^{(2)}), \quad X^{(1)}, X^{(2)} \stackrel{\text{iid}}{\sim} \mu. \quad (1.2)$$

The classical contraction argument does not work. Since $X = 0$ is always a solution, we cannot use the Wasserstein metric W_p . Since T_α can change the mean of the distribution, we cannot use the Zolotarev metric ζ_p .

1.1 Background

1.1.1 Empirical Distributions

Given a distribution μ and the number of particles n , we can sample

$$X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \mu.$$

The empirical distribution μ_n is

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}.$$

Definition 1 (Weak convergence of distributions). *Let (M, d) be a metric space. Let $\mathcal{M}$ be the Borel σ -algebra on M . We define the distribution on $(M, \mathcal{M})$. Let $C_b(M)$ be the set of real-valued, bounded, continuous functions on M .*

A sequence of distributions (μ_n) converges weakly to a distribution μ if for $f \in C_b(M)$, we have

$$\int f d\mu_n \rightarrow \int f d\mu.$$

Theorem 1 (Varadarajan). *The sequence μ_n converges weakly to μ almost surely.*

Remark. We need to study the rate of convergence, for simulation efficiency.

By Continuous Mapping Theorem, we can do arithmetic operations on the empirical distributions.

Proposition 1. *Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \mu$ and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$. Let $f : M \rightarrow M$ be a continuous function. Then*

$$\frac{1}{n} \sum_{i=1}^n \delta_{f(X_i)} \xrightarrow{w} f_{\#} \mu$$

almost surely.

We use the following particular cases

1. When $f(x, y) = x + y$, then $\frac{1}{n} \sum_{i=1}^n \delta_{X_i+Y_i} \xrightarrow{w} \mathcal{L}(X + Y)$ almost surely.
2. When $f(x, y) = \max(x, y)$, then $\frac{1}{n} \sum_{i=1}^n \delta_{\max(X_i, Y_i)} \xrightarrow{w} \mathcal{L}(\max(X, Y))$ almost surely.
3. when $f(x, y) = xy$, then $\frac{1}{n} \sum_{i=1}^n \delta_{X_i Y_i} \xrightarrow{w} \mathcal{L}(XY)$ almost surely.

Proposition 2. *Let (μ_n) be empirical measures such that $(\mu_n) \xrightarrow{w} \mu$ almost surely and the p -th moment of μ is finite. Then*

$$W_p(\mu_n, \mu) \rightarrow 0$$

almost surely.

Using the triangle inequality, we can show the following convergence result.

Proposition 3 (Convergence of Wasserstein distance). *Let $(\mu_n) \xrightarrow{w} \mu$ almost surely and $\nu_n \xrightarrow{w} \nu$ almost surely. Then*

$$W_p(\mu_n, \nu_n) \rightarrow W_p(\mu, \nu)$$

almost surely.

Proposition 3 provides an $O(n \log n)$ algorithm to compute the empirical Wasserstein distance

$$W_p(\mu_n, \nu_n) = \frac{1}{n} \inf_{i,j} \left(\sum_{i=1}^n |X_i - Y_j|^p \right)^{1/p} = \frac{1}{n} \left(\sum_{i=1}^n |\hat{X}_i - \hat{Y}_i|^p \right)^{1/p}, \quad (1.3)$$

where $(\hat{X}_1, \dots, \hat{X}_n)$ and $(\hat{Y}_1, \dots, \hat{Y}_n)$ are the sorted samples from μ_n and ν_n , respectively.

1.1.2 Random Distributions

Our conjectures include convergence of distributions. We need a way to generate random distributions. In particular, we need to generate random distributions with mean 1.

References

- [1] Michael Drmota. *Random trees: an interplay between combinatorics and probability*. Springer, 2009.
- [2] R Ludger. “The contraction method for recursive algorithms”. In: ().
- [3] Ralph Neininger and Ludger Rüschendorf. “A general limit theorem for recursive algorithms and combinatorial structures”. In: *The Annals of Applied Probability* 14.1 (2004), pp. 378–418.